

GULKANA, AK, USA

WMO: 702710

Lat: 62.159N

Lon: 145.459W

Elev: 476

StdP: 95.73

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26425

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.4	-35.7	-36.9	0.1	-33.5	-35.5	0.1	-32.3	11.7	0.4	10.6	0.8	0.5	310	0.758

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	25.7	14.6	23.6	13.8	21.6	13.0	15.7	23.7	14.6	21.9	13.6	20.1	3.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	9.4	16.9	11.3	8.8	16.2	10.3	8.3	15.5	45.1	24.0	42.0	21.9	39.4	20.0	19.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.2	8.0	-40.2	28.0	2.2	2.1	-41.8	29.5	-43.1	30.8	-44.4	32.0	-46.0	33.5		
(5)			-36.3	16.7	1.6	1.3	-37.4	17.7	-38.4	18.4	-39.3	19.1	-40.5	20.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.7	-18.4	-14.4	-9.5	0.4	7.7	12.8	14.4	12.2	6.7	-1.9	-13.9	-17.7
	DBStd	13.55	9.65	8.15	7.12	4.63	3.45	3.02	2.63	2.88	3.57	6.04	8.51	9.11
	HDD10.0	4609	881	682	603	287	88	7	1	11	106	369	717	859
	HDD18.3	7326	1139	916	861	537	328	168	125	192	348	627	967	1117
	CDD10.0	331	0	0	0	0	18	91	137	77	7	1	0	0
	CDD18.3	6	0	0	0	0	0	2	3	1	0	0	0	0
	CDH23.3	200	0	0	0	0	6	70	81	43	0	0	0	0
	CDH26.7	29	0	0	0	0	0	14	10	4	0	0	0	0
Wind	WSAvg	2.5	1.4	1.7	2.5	3.1	3.4	3.2	3.1	3.0	3.0	2.2	1.6	1.4
Precipitation	PrecAvg	279	12	12	8	6	18	40	45	41	38	24	16	20
	PrecMax	378	38	42	34	21	55	103	84	100	100	62	73	65
	PrecMin	134	0	0	0	0	0	7	12	1	2	3	0	0
	PrecStd	63	8	11	8	6	14	21	19	21	24	16	14	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.6	4.4	8.2	16.7	24.2	29.1	27.9	27.2	19.0	12.4	4.5	3.9
		MCWB	2.4	1.4	3.4	8.0	12.1	16.2	16.3	15.6	10.9	8.5	1.7	2.1
	2%	DB	2.5	2.5	4.9	12.5	21.1	25.2	25.6	24.4	16.7	9.3	3.1	2.1
		MCWB	1.0	0.3	1.1	5.6	10.4	13.8	15.0	14.5	9.8	5.4	1.1	0.6
	5%	DB	-0.3	0.6	3.1	10.1	18.5	22.8	23.5	21.5	14.8	7.6	1.4	-0.1
		MCWB	-1.8	-1.2	-0.2	3.9	9.0	12.8	14.4	13.4	9.1	4.3	0.0	-1.3
	10%	DB	-4.2	-1.9	1.3	8.3	15.9	20.5	21.6	19.1	13.3	6.1	-1.5	-3.9
		MCWB	-5.0	-3.4	-1.5	2.8	7.9	12.0	13.6	12.2	8.4	3.4	-2.4	-4.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	1.9	3.5	8.3	12.3	17.0	17.2	16.6	11.7	8.8	2.4	2.5
		MCDB	4.2	3.4	7.7	16.1	22.6	26.7	25.9	25.2	16.3	12.0	3.5	3.5
	2%	WB	1.2	0.5	1.5	6.0	10.9	14.8	15.8	15.1	10.6	6.1	1.2	0.7
		MCDB	2.7	2.3	4.4	12.1	20.1	23.6	23.6	22.8	15.2	8.4	2.8	1.9
	5%	WB	-1.7	-1.1	0.1	4.2	9.6	13.4	15.0	13.9	9.8	4.7	0.0	-1.4
		MCDB	-0.2	0.7	2.9	9.3	17.1	21.1	22.2	20.4	14.1	7.1	1.5	-0.1
	10%	WB	-4.7	-3.3	-1.3	3.0	8.4	12.4	14.0	12.8	8.9	3.5	-2.3	-4.5
		MCDB	-3.7	-1.7	1.4	7.8	15.4	19.3	20.5	18.1	12.6	5.9	-1.4	-3.5
Mean Daily Temperature Range	5% DB	MDBR	8.2	10.9	13.9	12.3	12.6	12.5	11.5	11.7	11.0	9.0	8.4	8.3
		MCDBR	8.9	10.1	12.0	14.8	18.1	16.8	16.2	16.9	14.0	9.2	8.0	9.5
		MCWBR	7.8	8.4	8.8	8.8	8.9	7.7	7.4	8.4	8.2	6.2	6.9	8.6
	5% WB	MCDBR	8.7	10.0	10.8	14.1	15.8	15.1	14.4	14.8	12.3	8.8	7.7	9.0
		MCWBR	7.7	8.5	7.9	8.5	8.1	7.2	6.8	7.5	7.5	6.3	6.7	8.2
Clear-Sky Solar Irradiance	taub	0.207	0.263	0.272	0.356	0.356	0.368	0.363	0.358	0.303	0.279	0.215	0.188	
	taud	2.050	2.184	2.246	2.094	2.342	2.361	2.402	2.420	2.556	2.449	2.057	1.908	
	Ebn at Noon	492	703	846	819	862	855	850	823	821	700	458	290	
	Edh at Noon	34	64	87	127	108	109	102	92	66	50	33	21	
All-Sky Solar Radiation	RadAvg	0.26	1.00	2.76	4.60	5.43	5.45	4.92	3.83	2.47	1.13	0.37	0.12	
	RadStd	0.03	0.11	0.22	0.25	0.30	0.41	0.33	0.34	0.26	0.14	0.05	0.01	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+2
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page